

NANYANG JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATIONS  
Higher 2

CANDIDATE  
NAME

CLASS

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**BIOLOGY**

**9648/02**

Paper 2 Core Paper

**September 2011**

Additional Materials: Answer Paper

**2 hours**

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**READ THESE INSTRUCTIONS FIRST**

Write your name and CT on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use soft pencil for any diagrams, graphs or rough working.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

**Section A**

Answer **all** questions.

**Section B**

Answer any **one** question.

At the end of the examination, fasten all your work securely together.  
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |  |
|--------------------|--|
| Section A          |  |
| 1                  |  |
| 2                  |  |
| 3                  |  |
| 4                  |  |
| 5                  |  |
| 6                  |  |
| 7                  |  |
| Section B          |  |
| Total              |  |

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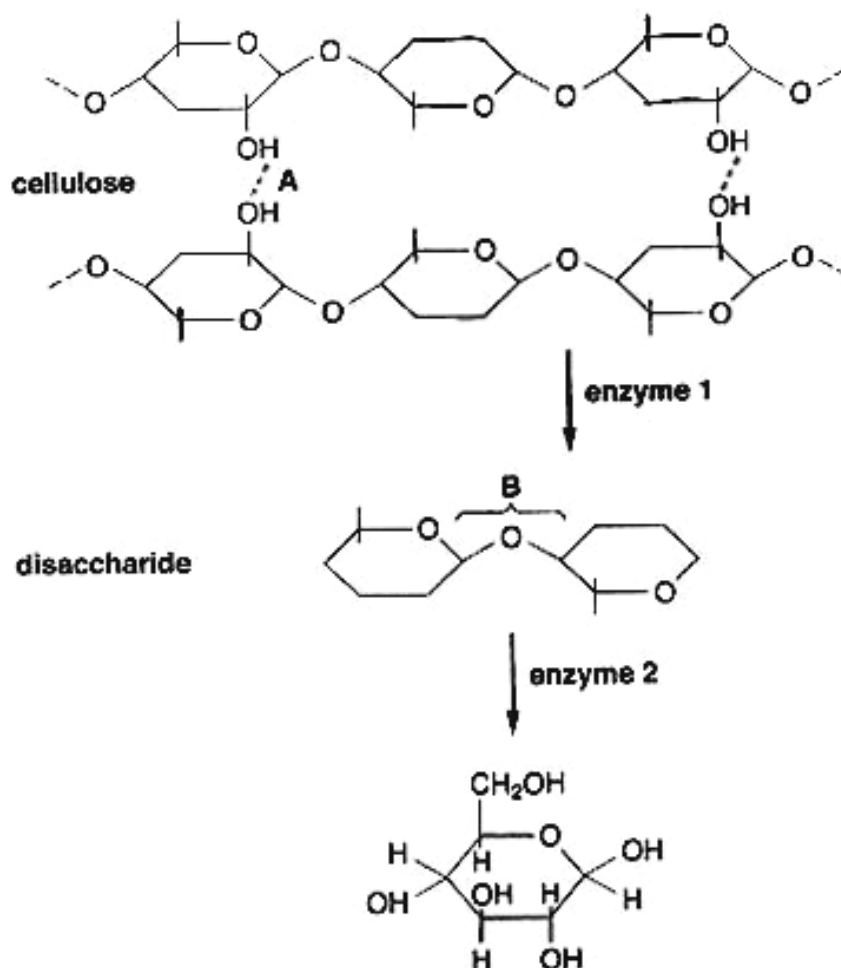
This document consists of **15** printed pages and **2** blank pages.

**[Turn over**

## Section A

Answer **all** the questions in this section.

- 1 In the search for new biofuels, research has been done into the digestion of wood waste by fungi. The cellulase enzymes produced by the fungi break cellulose into sugars. These sugars can then be converted into ethanol, a biofuel. **Fig. 1.1** shows the stages in this digestion process.



**Fig. 1.1**

- (a) Name bonds **A** and **B** shown in **Fig. 1.1**.

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 [2]

- (b) Describe how bond **B** is broken in the digestion of the disaccharide.

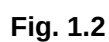
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 [1]

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**Fig 1.2** shows a molecules commonly found in cell membranes.



[2]

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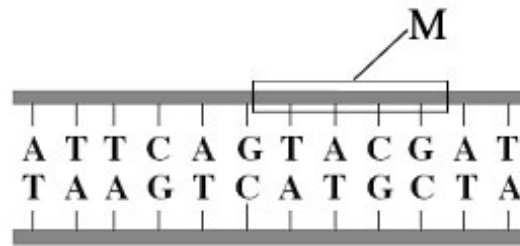
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[Total: 10]

- 2 **Fig 2.1** shows part of a molecule found in a species of bacterium.



**Fig. 2.1**

- (a) Name the **two** components that make up the region labeled **M**.

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[2]

- (b) Scientists calculated the percentage of different bases in the DNA from a species of bacterium. They found that 14% of the bases were guanine.

- (i) What percentage of the bases in this species of bacterium was adenine?

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[1]

- (ii) In another species of bacterium, 29% of the bases were guanine.

Explain the difference in the percentage of guanine bases in the two species of bacterium.

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[2]

**Fig. 2.2** shows a DNA base sequence. It also shows the effect of two mutations on this base sequence. **Fig. 2.2** shows DNA triplets that code for different amino acids.

| <b>A</b>                     |   |   |   |   |   |   |   |   |   |   |   |   |
|------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|
| Original DNA base sequence   | A | T | T | G | G | C | G | T | G | T | C | T |
| Amino acid sequence          |   |   |   |   |   |   |   |   |   |   |   |   |
| Mutation 1 DNA base sequence | A | T | T | G | G | A | G | T | G | T | C | T |
| Mutation 2 DNA base sequence | A | T | T | G | G | C | C | T | G | T | C | T |

| <b>B</b>           |            |
|--------------------|------------|
| DNA triplets       | Amino acid |
| GGT, GGC, GGA, GGG | Gly        |
| GGT, GTA, GTG, GTC | Val        |
| ATC, ATT, ATA      | Ile        |
| TCC, TCT, TCA, TCG | Ser        |
| CTC, CTT, CTA, CTG | Leu        |

**Fig. 2.2**

- (c) Complete **Fig. 2.2A** to show the sequence of amino acids coded for by the original base sequence.

[1]

- (d) Some mutations affect the amino acid sequences while others do not. Using the information in **Fig. 2.2**, explain

- (i) why mutation 1 has no effect on the amino acid sequence;

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[1]

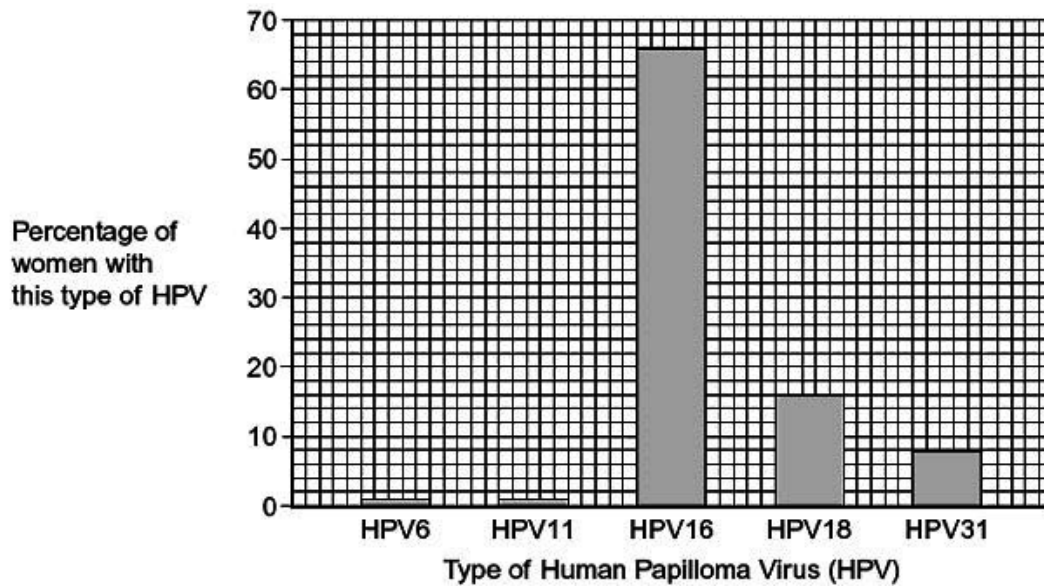
- (ii) how mutation 2 could lead to the formation of a non-functional enzyme.

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[3]

[Total: 10]

- 3 Cervical cancer occurs in the cervix. The cervix is the opening of the uterus. Scientists investigated the link between cervical cancer and infection with some types of Human Papilloma Virus (HPV). **Fig. 3.1** shows the frequency of five different types of HPV in women who had cervical cancer.



**Fig. 3.1**

- (a) A local newspaper published an article about cervical cancer with the headline “HPV causes cervical cancer.”

Using data from **Fig 3.1**, comment if this claim may be supported.

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[2]

A study published in the 28<sup>th</sup> edition of *AIDS* has shown that HIV-positive women are more likely to carry multiple strains of human papilloma virus (HPV). It has been proven that early intervention by antiviral drugs can prevent HIV.

- (b) Suggest an explanation for the high incidence of HPV in HIV patients.

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[2]

- (c) Suggest and explain the possible modes of action of the antiviral drugs in halting the reproductive cycle of HIV.

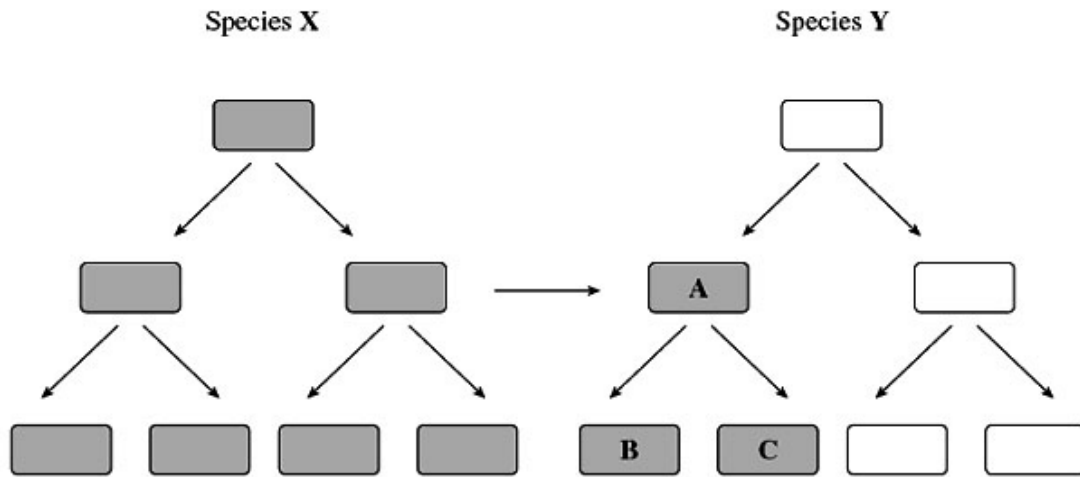
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[3]

Species **X** and **Y** are bacteria. **Fig. 3.2** shows gene transfer between bacteria in these two species. The bacteria that are shaded are resistant to the antibiotic penicillin.



**Fig. 3.2**

**(d)** Using **Fig 3.2**,

**(i)** explain how bacterium **A** becomes resistant to penicillin;

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[4]

**(ii)** explain why bacteria **B** and **C** are resistant to penicillin.

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[3]

[Total: 14]

- 4 Sex hormone estrogen promotes mammary epithelial cell proliferation by stimulating the expression of genes encoding cell cycle regulatory proteins. Estrogen receptors (ER $\alpha$ ) are intracellular receptors of estrogen. Estrogen-receptor complexes bind to estrogen response elements (ERE) in enhancers. Estrogenic control of the cell cycle occurs by increasing the expression of *cyclin* and *myc C* which promote cell cycle progression.

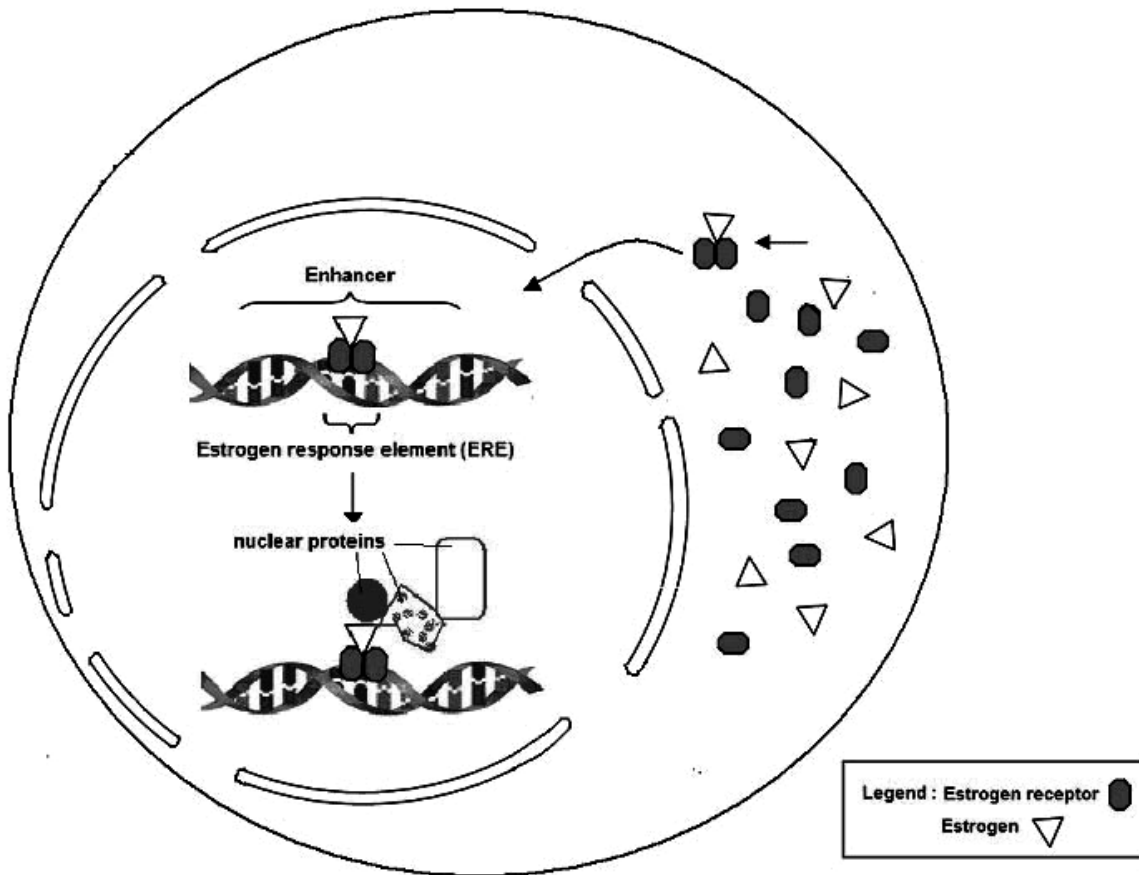


Fig 4.1

- (a) With reference to the Fig 4.1,  
(i) State the level at which gene expression is regulated by estrogen;

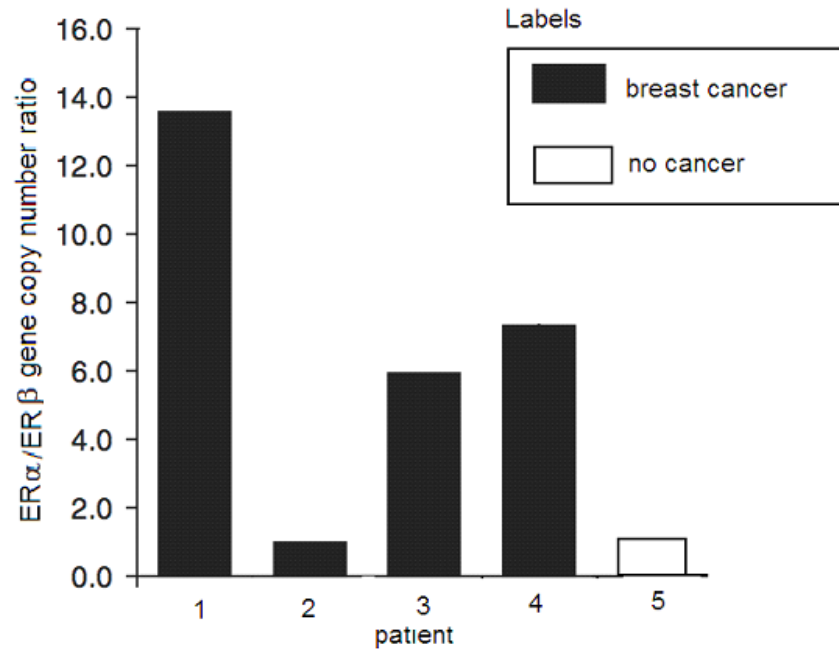
[1]

- (ii) describe and explain how estrogen regulates gene expression;

[4]



There are two isoforms of estrogen receptors: ER $\alpha$  and ER $\beta$  encoded by different genes. Genomic DNA samples taken from women suffering from breast cancer were analysed for the gene copy number present for the ER $\alpha$  gene.



**Fig 4.2**

- (b) With reference to the **Fig 4.2**, describe the effect different ER $\alpha$  gene copy number on the women.

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[2]

- (c) Suggest an explanation as to why patient 2 developed cancer despite a low copy number.

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[1]

- (d) Suggest how amplification of the ER gene contributes to breast cancer.

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[2]

[Total: 10]

- 5 Coat colour in cats is determined by a sex-linked gene with two alleles, black and ginger. When black cats are mated with ginger cats, the female offspring are always tortoiseshell, their coats showing a mottling of small black and ginger patches, while the male offspring always have the same coat colour as their mothers.
- (a) Using suitable symbols, construct genetic diagrams to explain these results.

[6]

- (b) In a litter of kittens, half the number of females were tortoiseshell and the other half were black; half the number of males were black and the other half were ginger. State and explain what the colours of the parental male and female were.

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[4]

[Total: 10]

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- 6 (a) State the roles of glycoproteins, carrier proteins and cholesterol in the cell surface membrane of an animal cell.

*glycoproteins*

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*carrier proteins*

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*cholesterol*

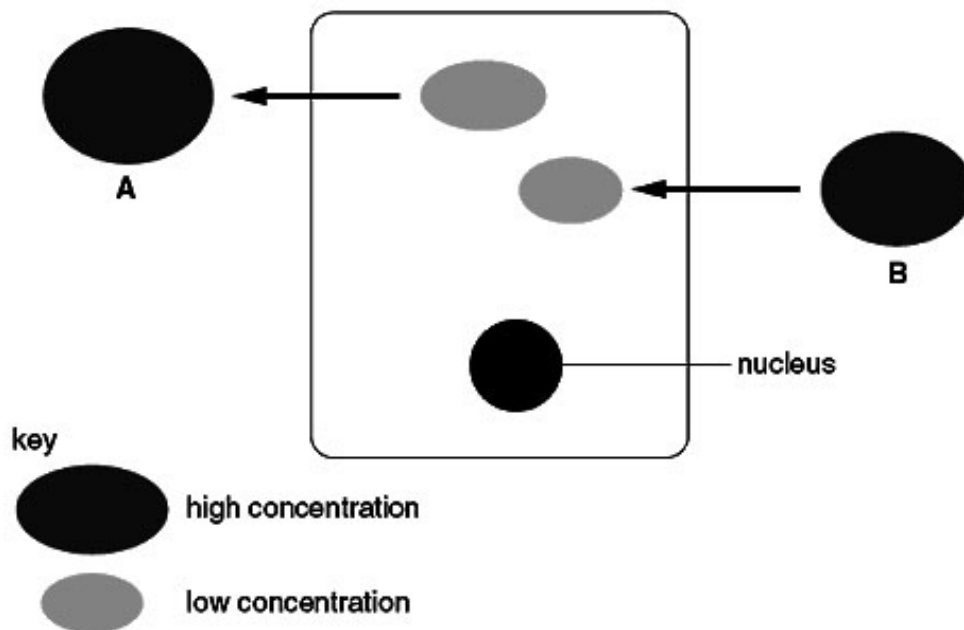
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[4]

**Fig 6.1** shows an animal cell indicating the concentrations and direction of movement of an ion (**A**) and a non-polar molecule (**B**) on either side of the cell surface membrane.



**Fig 6.1**

- (a) With reference to **Fig 6.1**, explain how **A** and **B** cross the cell surface membrane of the cell.

**A**

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.....

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**B**

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[4]

- (b) Describe how particles, such as bacteria, are taken up by phagocytes.

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[2]

- (c) Phagocytes contain many lysosomes. State the function of lysosomes in phagocytes.

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[1]

[Total: 11]

- 7 In a recent scientific expedition to study a vulnerable species of the cat family (*Felidae*), it was found that this species seemed to be heading for extinction. This species once spanned the globe, but its range is now limited to a few pockets in Africa. It was suggested that the species had somehow lost its genetic variation. To verify this suggestion, electrophoresis was used to examine the genetic variability in populations of this species. The variability was very low when compared with most populations of other mammals. If one were to select two individuals of this species at random, the chance that both being genetically very similar would be very high.

(a) Explain the following terms:

(i) species;

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.....  
[2]

(ii) genetic variation.

.....  
.....  
[1]

(b) Explain the impact that low genetic variability in a population of this species has on species survival.

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.....  
[2]

In the wild, this feline species possessed impressive skills both as a runner and hunter. However, it was discovered that there was variation in the number of spontaneous movement (i.e. tweaking movement) which occurred when the different individuals of the species run. The higher the amount of spontaneous movement, the slower the running speed of the individuals. **Fig. 7.1** shows the number of spontaneous movement (activity score) recorded when the species was placed in standardized laboratory conditions.

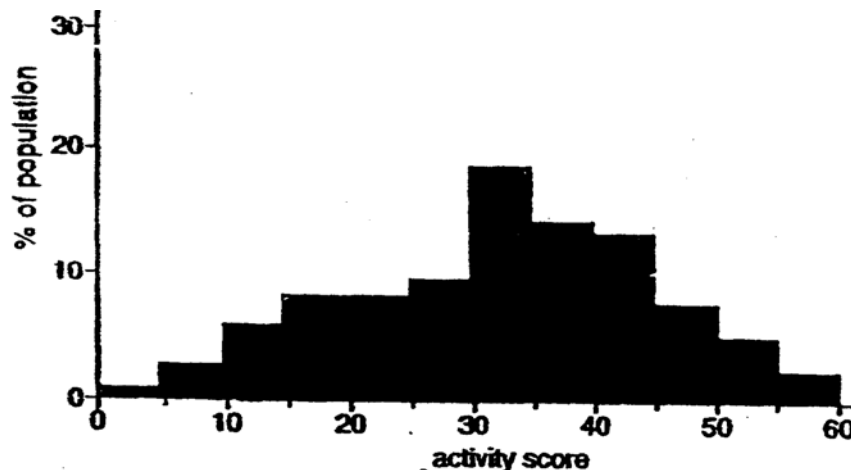


Fig. 7.1

(c) Using the information from **Fig. 7.1**, explain the type of variation shown.

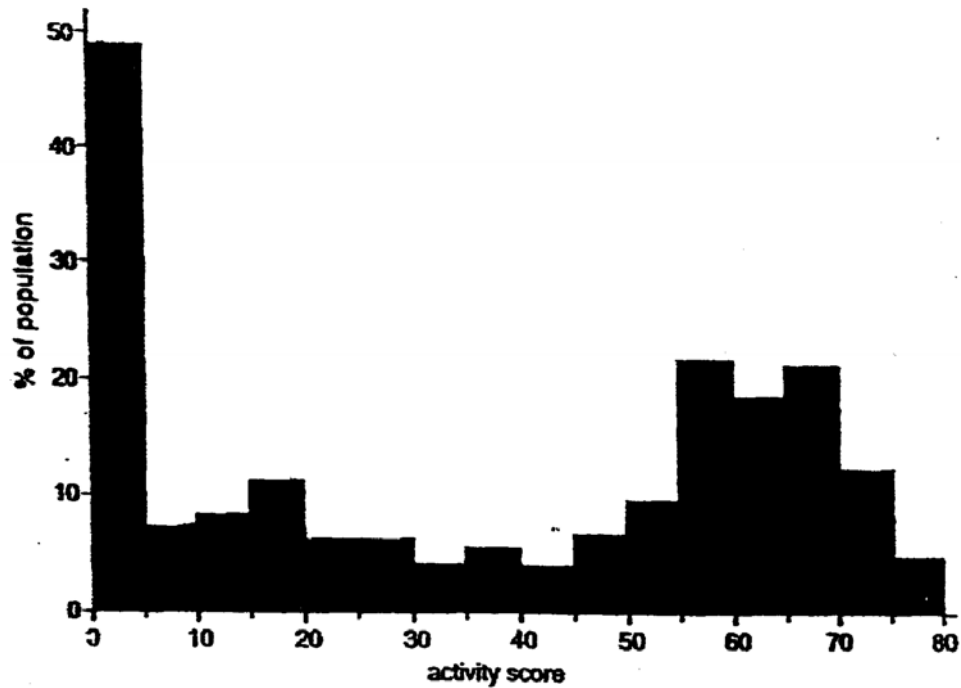
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[2]

The scientists then took the individuals with a low activity score and allowed them to breed among themselves. They did the same with those individuals showing high activity score. The result after 20 generations is shown in **Fig. 7.2**.



**Fig. 7.2**

(d) With reference to **Fig. 7.2**,  
(i) state the type of selection.

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[1]

(ii) suggest what may happen if the high activity score species produce by the breeding experiment are now introduced to the low activity score species.

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[2]

(e) Being hunters, aggression is a trait observed in this feline species.

(i) Suggest how aggression probably evolved in this species.

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[2]

(ii) Explain why the feline population is the smallest unit that allows for the evolution of aggression.

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[3]

[Total: 15]



**Section B**

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8** (a) Describe the structural features of **two** named nucleic acids involved in protein synthesis. [6]
- (b) Compare the two processes that have to take place in order for protein synthesis to occur in prokaryotes. [8]
- (c) Outline how the rate of protein synthesis in eukaryotes may be controlled. [6]
- 9** (a) Explain how the light independent reaction is important in photosynthesis. [8]
- (b) Describe how Calvin cycle differs from Krebs cycle. [6]
- (c) State the similarities between ATP production in mitochondria and chloroplasts and suggest why these similarities exist. [6]